# TeruBot: Materializing Collective Emotions Through an Embodied Robotic Metaphor


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**Figure 1: TeruBot, a community-based social robot.**


## Abstract

Emotional sharing fosters social bonds, yet in shared environments such as workplaces, collective emotional states often remain invisible, contributing to discomfort or disconnection. We present TeruBot, a hybrid robot–application system for expressing and visualizing collective emotions. Inspired by the Japanese Teru Teru Bōzu, a handmade puppet believed to bring good weather, TeruBot reframes “good weather” as a tangible metaphor for emotional and social well-being.

TeruBot combines a companion robot with a lightweight mobile application through which individuals share their emotional state. These inputs are aggregated into a shared “emotional weather”




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expressed by the robot in a communal space. By rendering collective emotional states as a tangible metaphor, TeruBot explores how playful and ambiguous systems can support emotional awareness and social connection within communities.

## CCS Concepts

• **Human-centered computing** → **Human–robot interaction**; **Affective computing**; *Collaborative and social computing*.

## Keywords

collective emotions, social robots, affective atmospheres, human–robot interaction, community wellbeing

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## 1 Introduction & Related Work

Emotions shape how people experience everyday environments by influencing well-being and a sense of belonging within spaces and social interactions [9]. Beyond individual feelings, emotions are also experienced collectively as "affective atmospheres": indeterminate forces that emerge through social interaction in shared spaces [1]. These collective emotional states are often sensed implicitly but rarely articulated, remaining largely invisible despite their influence on how communities function.

In shared environments such as workplaces and educational spaces, unspoken collective emotions can contribute to discomfort, misunderstanding, or emotional distance. Prior work shows that perceived psychological distance in organizations leads to disconnection [12], unresolved emotional climates impair communication and team cohesion [6], and unregulated collective emotions can cascade into negative feedback loops that alienate community members [5]. Despite their social impact, such emotional climates are rarely made perceptible or open to shared reflection.

Within HCI, research on affective technologies has increasingly engaged with emotional data [19, 21], but has largely focused on individual-level monitoring [2], reflection [7], or regulation [18]. Although collective emotional states have been modeled computationally as non-linear emotional fields or clusters [10, 16], their representation is typically confined to screen-based dashboards or visualizations, leaving limited exploration of how such states might be materialized in shared physical space [13, 15].

This individual-centered orientation extends to social robots designed for emotional interaction. Many such systems support emotional expression and well-being through one-on-one companionship, particularly in mental health contexts [17]. Rather than prioritizing precise emotion recognition, affective robots often convey emotional states through expressive movement, posture, or tone, as users have been found to value consistency over emotional accuracy [11, 19].

More recent work has begun to explore robots and embodied artifacts as socio-spatial catalysts that operate at the level of shared environments rather than individuals. Situated artifacts in semi-public spaces can shape collective attention and social dynamics [4], while systems such as MirrorBot externalize group-level behavior to support shared awareness of presence [8]. Playful and intentionally ambiguous representations, as demonstrated by projects like F.I.S.H. [20], suggest that openness and interpretability can lower social barriers and encourage communal reflection.

Building on these directions, we introduce TeruBot, a companion robot designed to reflect the emotional climate of a community. Paired with a lightweight mobile application, TeruBot aggregates individual emotional inputs into a shared physical representation. Rather than aiming for precise emotion recognition, the system adopts an intentionally ambiguous design that invites interpretation and reflection on collective emotional states. Through this work, we contribute an embodied, metaphor-driven approach to materializing collective emotion in shared spaces, together with its implementation and preliminary exploratory findings on user perceptions.

## 2 Design & Implementation

### 2.1 Design Rationale

TeruBot externalizes collective affect through non-verbal, embodied cues combining expressive eye animations and simple head gestures. The system adopts a lightweight multimodal vocabulary: the eyes convey atmospheric qualities through openness, blink rhythm, and symbolic elements, while head movements reinforce expression through posture and motion. The visual design draws on the "baby schema" (*Kindchenschema*), whose large eyes and rounded features elicit caregiving responses and increase the social appeal of inanimate objects [14].

Positive states are represented by wide-open eyes with star-like highlights and a gentle left-right head tilt, while low or sad states use half-closed eyes with water droplets and a slow downward pitch. Head gestures alone convey interactional signals, including a brief side tilt for questioning and nodding or shaking for affirmation or negation. These stylized, metaphorical behaviors encourage interpretation rather than fixed emotional labeling. Figure 3 summarizes the mapping between aggregated emotional states, weather metaphors, and the robot's multimodal expressions.

**(a) Happy mood made on Python with Pygame**

**(b) Sad mood with top eyelids down and tears, with Pygame**

**Figure 2: Mood expressions created with Pygame**

## 3 Implementation

This section describes the implementation of TeruBot, which consists of a physical robot and an accompanying mobile application, connected through a backend that manages communication and data flow between them.

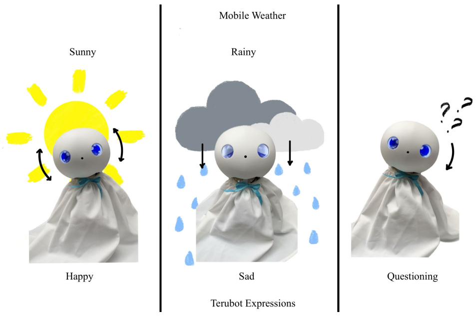


**Figure 3: App weather and Terubot expression mapping**

## 3.1 Hardware Overview

This subsection describes the hardware components of TeruBot, including its processing unit, sensing elements, actuation for movement, and expressive eye displays.

*3.1.1 Robotic Architecture and Movement.* TeruBot is built around a Raspberry Pi as the main processing unit. It runs a local large language model (Qwen 2.5–1.5B) to process voice input, extract emotional keywords, and select the robot's expressive behavior. A Logitech C270 webcam in the nose handles presence and gaze detection, while its integrated microphone captures speech.

The Raspberry Pi communicates via serial with three ESP32 microcontrollers. Two Waveshare 1.28-inch ESP32-S3 round displays act as eyes, and a third ESP32 drives several MG996R servo motors for head movements such as nodding, shaking, and tilting.

Speech from the microphone is converted to text via a speech-to-text model. The LLM analyzes this text for semantic sentiment (e.g., joy or sadness), which triggers the robot's response. These emotional states are mapped to behavioral presets: for example, "positive" text elicits nodding and bright eye animations, while "negative" text elicits a slow downward pitch or drooping eyes.

*3.1.2 Expressive Eye Hardware and Animation.* TeruBot's face is composed of two animated eyes displayed on separate screens. One screen is controlled by the Raspberry Pi, which communicates via a serial connection with the second screen in a master–slave configuration to maintain synchronized behavior.

Rather than relying on pre-rendered frames, eye expressions are generated through procedural animation. Parameters such as eyelid curvature, iris position, blink speed, and eye openness are continuously modulated to produce smooth and dynamic transitions between expressive states.

The microcontrollers communicate through a serial protocol to coordinate animations across both eyes. This architecture enables lightweight and synchronized visual behaviors without relying on explicit emotion labels or high-level facial models.

## 3.2 Mobile Application

The mobile application enables individuals to submit brief emotional check-ins that contribute to the robot's collective emotional state. It supports participation independently of physical proximity to the robot and allows asynchronous interaction within the community.

*3.2.1 Collective Emotional State Visualization.* Collective emotional states are represented using a weather-based abstraction inspired by the *Teru Teru Bōzu.* Individual emotional inputs are aggregated at the community level and mapped to coarse weather conditions (e.g., sunny, cloudy, stormy). This approach allows the overall emotional climate to be shared without exposing individual inputs or personal details.

*3.2.2 Emotional Check-in Interface.* To support regular use, the emotional check-in is designed to be brief and low-effort. Rather than relying on free-text input, the interaction is structured around four dimensions: general mood, energy level, stress or mental load, and sense of social connection. Users respond to a short set of questions corresponding to these dimensions, without explicitly labeling emotions or writing text. Specifically, the interface prompts users with the following questions:

(1) *How are you feeling today? (general mood)*
(2) *What is your current energy level? (energy)*
(3) *How connected do you feel with others? (social connection)*
(4) *Do you feel stressed at the moment? (stress/mental load)*

The resulting input is synthesized into a compact representation that contributes to the shared emotional state reflected by the robot.

*3.2.3 Community Channel.* The application includes a shared community channel where users can post short, anonymous positive messages. The channel is intentionally collective rather than conversational, avoiding direct messaging to reduce interpersonal pressure while supporting a sense of shared presence within the community.

## 3.3 Software Architecture and Data Flow

TeruBot is supported by a backend that connects the mobile application with the physical robot, handling authentication, storage, and aggregation of emotional inputs. The system follows a decoupled client–server architecture, allowing components to communicate while remaining loosely coupled.

The backend is implemented using Node.js and Express.js and exposes a RESTful API for emotional check-ins. Inputs are stored in a MongoDB database and validated using Mongoose, while authentication relies on JWT tokens and BCrypt. Emotional inputs are aggregated over time to update the robot's current expressive state.

# 4 Preliminary Study

We conducted a formative user study to evaluate the usability, perceived quality, and trust-related aspects of the system. The study focuses on two interaction layers: (1) user–app interaction, (2) user–robot interaction. Eight participants (N = 8) were recruited to test the mobile application and interact with the robot in a controlled setting. Participants were introduced to the system's concept and then asked to freely explore the app, complete an emotional check-in, and observe the robot's expressive behavior. After the interaction, they completed a *Godspeed Questionnaire*[3], using 5-point Likert scales (1 = strongly disagree, 5 = strongly agree), followed by an open-ended comment question.

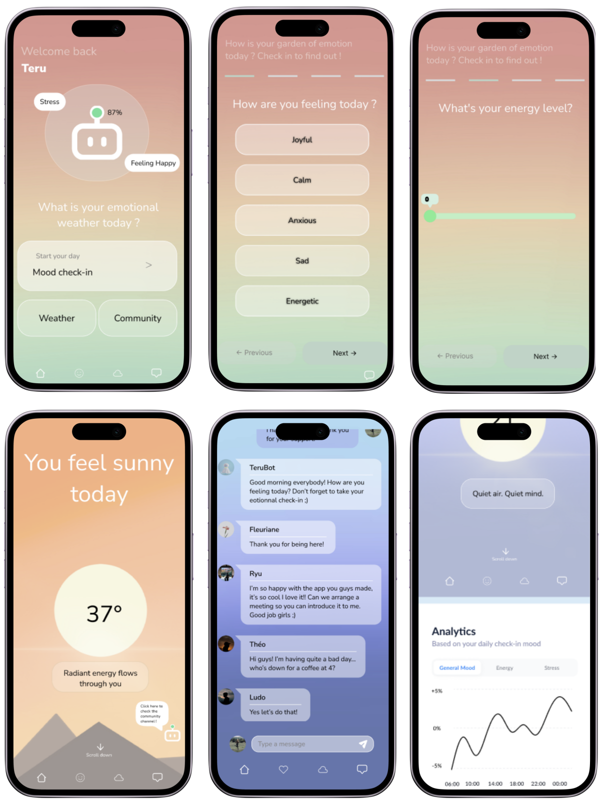


**Figure 4: Mockups of the mobile application interface. From left to right, top to bottom: the home dashboard, mood check-in screen, energy level slider, mood summary visualization, community chat with the TeruBot assistant, and an analytics page displaying emotional trends.**

## 4.1 User–App interaction

To assess the usability and experiential quality of the application, participants completed a short questionnaire composed of five non-leading Likert-scale items (1–5). These items evaluated overall experience, clarity of the pages and features, comfort with the emotional check-in, ease of emotional reflection, and willingness to use the app regularly. An optional open-ended question allowed participants to provide additional feedback and qualitative impressions.

## 4.2 User–Robot interaction

To evaluate trust and emotional acceptance of the robot as a companion, participants rated three trust-related statements on a 5-point Likert scale. These statements assessed their comfort sharing emotions with the robot, the perceived trustworthiness of the robot, and whether the robot's presence felt more supportive than using the app alone. To assess the quality of the interaction between users and the robot, participants rated the following statements on a 5-point Likert scale: 1. The robot's behavior was easy to understand. 2. The robot's movements and expressions felt appropriate to the situation. 3. Interacting with the robot felt natural and engaging. 4. The robot enhanced my overall experience with the system. These questions were designed to capture clarity, expressiveness, coherence, and perceived added value of the robot within the interaction.

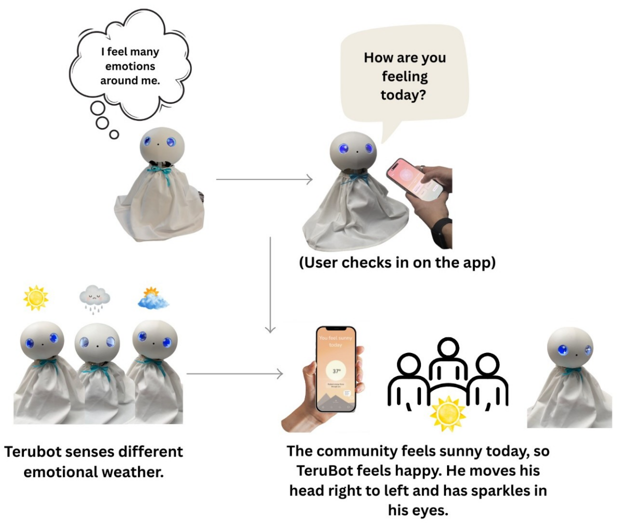


**Figure 5: Interaction scheme of TeruBot, from individual emotional input through the mobile application to the robot's collective emotional expression.**

## 4.3 Results

Participants reported positive experiences with the application. Ratings for overall experience ranged from 3 to 4. Clarity of pages and features was rated between 4 and 5. Comfort with the emotional check-in questionnaire received the highest ratings, with all participants selecting 5. Ratings related to emotional reflection ranged from 3 to 4. Willingness to use the app regularly was rated between 4 and 5. Participants rated the robot's behavior as easy to understand, with scores ranging from 4 to 5. Appropriateness of the robot's movements and expressions ranged from 3 to 5. Ratings for how natural and engaging the interaction felt ranged from 3 to 4. All participants rated the robot as enhancing the overall experience with the system, with a score of 5. Participants provided open-ended comments highlighting several observations, including appreciation of the app's visual design and colors, questions about flexibility in the emotional check-in process, and potential alternative uses of the community channel.

## 5 Discussion and Conclusion

The results suggest that both the application and the robot were perceived as usable and engaging, with particularly high comfort reported for emotional input and strong perceived added value. While emotional reflection scores were slightly lower, this may indicate opportunities to further support deeper reflection through interaction design. Participants also spontaneously interacted with the robot, such as touching or stroking its head, revealing unanticipated behaviors that highlight the importance of embodied interaction in emotional systems. TeruBot demonstrates how collective emotions can be materialized through an embodied and culturally inspired robotic metaphor. Collective emotion is operationalized through the aggregation of explicitly declared check-ins, supporting traceability and interpretive clarity, though limiting the model to declared affective states. While the community channel is not currently aggregated, integrating such discourse could provide a more layered, dialogical view of collective emotion. Acknowledging these dynamics highlights tensions between structured input and emergent expression, situating the system within broader debates on emotional technologies in professional environments. Future work will explore richer touch interactions and the potential integration of conversational input to deepen collective emotional reflection.

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